SIGHTLINE SCHEMATIC · STEP-BY-STEP GUIDE 2 OF 3

Revit: build the bowl mass family

Download the Revit package and run the Dynamo graph in a Generic Model family to create the bowl mass.

YOU NEED

- Revit 2024 with Dynamo 2.17 (installed with Revit, no extra packages)
- The bowl you want, open in Sightline Schematic

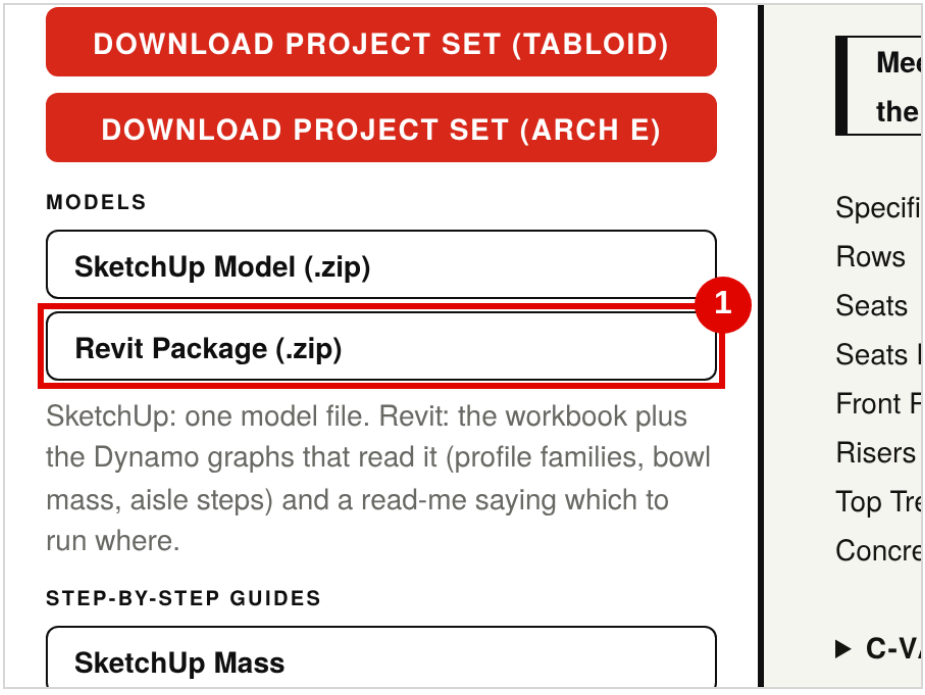
YOU GET

- One family holding the whole bowl: every stand extruded along the point-of-focus boundary, corners included
- One subcategory per stand, for colours and visibility
- Mass category, so Revit can make Mass Floors

STEPS

1 Download the Revit package

In Sightline Schematic, under **Deliver › Models**, click **Revit package (.zip)**. The browser saves `Stadium_Bowl_Revit_R2024.zip`.



Deliver › Models. (1) Revit package (.zip): the workbook, the Dynamo graphs and these guides.

2 Unzip it to a folder you keep

Extract the zip to the project folder, for example **…\Stadium\Sightline**. Dynamo reads the workbook from that folder, so do not run it from inside the zip.

Stadium_Bowl_Revit_R2024.zip		
XLSX	**Stadium_Bowl_Excel_YYYYMMDD_hhmmss.xlsx**	bowl data, input 1 of every graph
DYN	**Stadium_Bowl_Profile_R2024.dyn**	profile families
DYN	**Stadium_Bowl_Mass_Revit_R2024.dyn**	bowl mass family
DYN	**Stadium_Bowl_Aisle_Step_R2024.dyn**	precast aisle steps
TXT	**Stadium_Bowl_Aisle_Step_Family_Spec.txt**	aisle step family
PDF	**Stadium_Bowl_Mass_Revit_R2024_Guide.pdf**	this guide
PDF	**Stadium_Bowl_Profile_R2024_Guide.pdf**	profile guide
TXT	**READ ME.txt**	which graph to run where

The workbook name carries the date and time you downloaded it, so a new download never overwrites an older bowl.

3 Start a new Generic Model family

File › **New** › **Family…**

- Pick **Metric Generic Model.rft** (in **…\Family Templates\English**) or **Generic Model.rft** (in **…\English_I**). Units do not matter: the graph builds in true size.
- Do not use a Conceptual Mass template (Mass.rft): it cannot hold the extrusions and revolves the graph makes, and the graph stops with a message.
- Save the family now, as **Stadium_Bowl_Mass_Revit_R2024.rfa**.

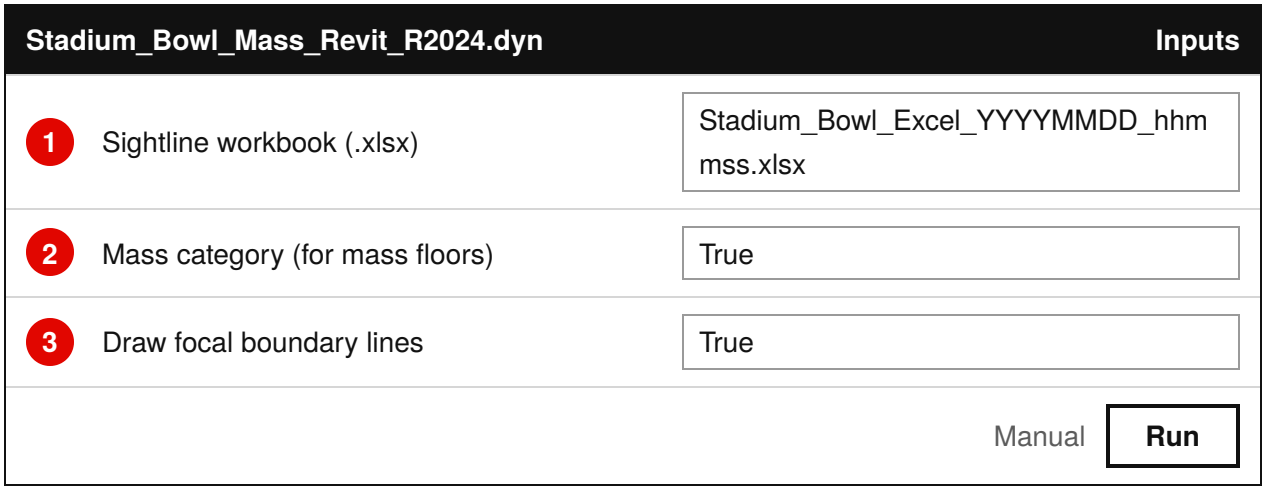
4 Open the graph in Dynamo

Manage › **Visual Programming** › **Dynamo**

In Dynamo, **File › Open** and pick **Stadium_Bowl_Mass_Revit_R2024.dyn**. Set the run mode at the bottom left to **Manual**.

Or use **Manage › Dynamo Player**, browse to the unzipped folder and click the graph: Player shows the same three inputs.

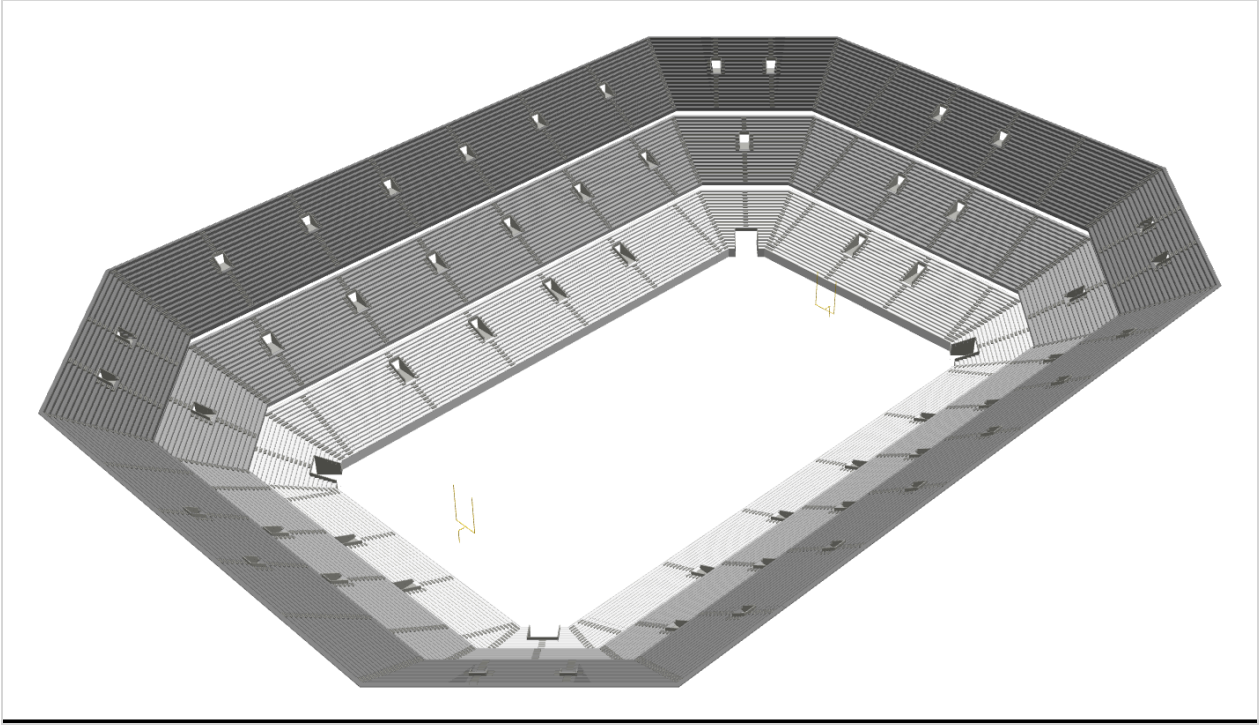
5 Set the inputs and run



Schematic of the graph inputs (numbered as in the graph). Click Browse on input 1 and pick the workbook from the unzipped folder.

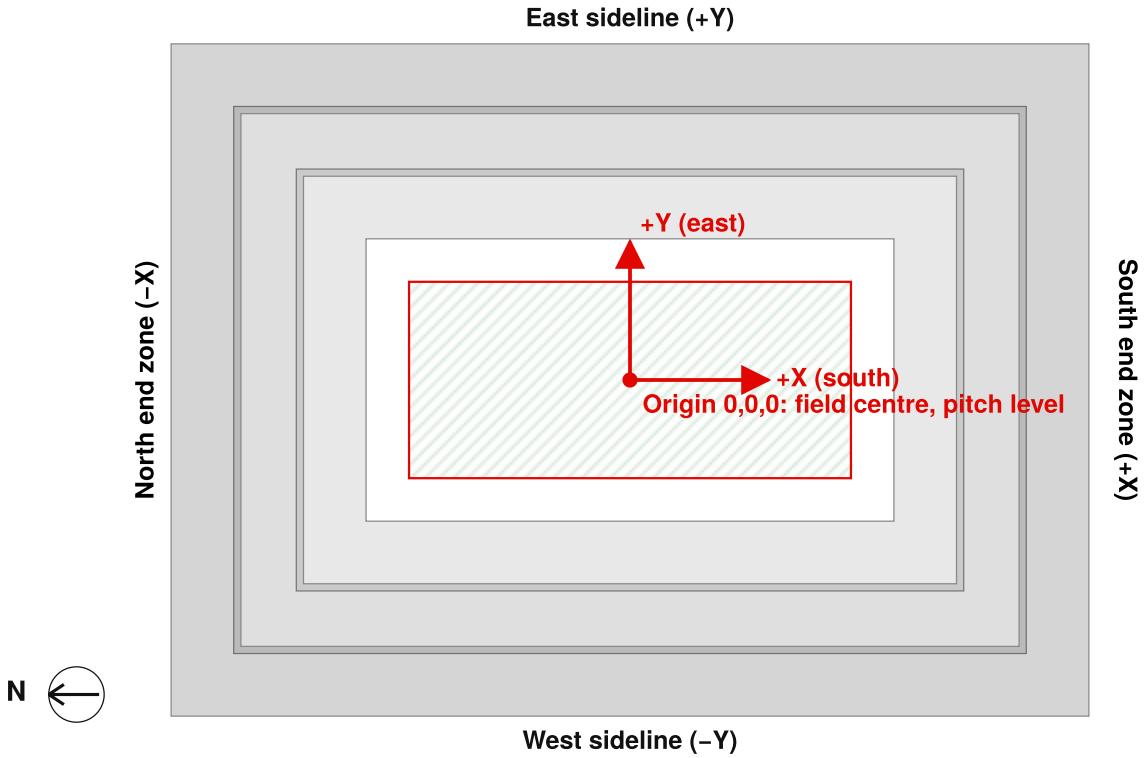
- **Input 2 True** changes the family category to **Mass**, which you need for Mass Floors. Leave it False to keep a Generic Model.
- **Input 3 True** draws the point-of-focus boundary as model lines: handy to check the bowl against the field.
- Click **Run**. The **Result** node lists the stands built (extrusions, revolves, corner plates) and the category. Running again replaces the bowl; it never stacks a second copy.

6 Check the family



Expected result in the family 3D view: the bowl as Sightline shows it in its 3D view.

- **Create › Properties › Family Category and Parameters** shows **Mass** (if input 2 was True).
- **Manage › Object Styles** lists one subcategory per stand (Sightline T1 E, Sightline T1 SEZ and so on), with a colour per tier.
- The family origin (where the Center reference planes cross) is the field centre at pitch level.



Family and project coordinates: origin at the field centre, north end zone on −X, East sideline on +Y. The red line is the point-of-focus boundary (input 3).

---

7 Load it into the project and place it

Create › **Family Editor** › **Load into Project**

- As a Mass: place it with **Massing & Site › Place Mass** (turn on **Show Mass** if Revit asks). As a Generic Model: **Architecture › Component › Place a Component**.
- Place it in a plan at the level of the pitch. To land it on the project origin, make the **Project Base Point** visible (Visibility/Graphics › Site), then **Move** the bowl from the crossing of its Center reference planes to the base point.
- Keep the bowl on the project origin if you will also run **Stadium_Bowl_Aisle_Step_R2024.dyn**: it places the aisle steps in the same coordinates.

---

8 Mass Floors (optional)

Select the bowl › **Modify | Mass** › **Model** › **Mass Floors**

Tick the levels to cut. Revit reports gross floor area per level from the bowl solid. This works only when the family category is Mass (input 2 True).

IF THE GRAPH STOPS

Message in the Result node	What to do
Run this in a family, not the project	You ran it with a project open. Do step 3 first and run it in that family.
This is a conceptual mass family	Start again from Metric Generic Model.rft and set input 2 to True.
Pick the workbook you exported from Sightline	Input 1 is empty or points at the zip. Browse to the .xlsx in the unzipped folder.
Tier … does not join up	The workbook is from an older Sightline version. Download the Revit package again.
Python errors on every node	Dynamo must run the Python nodes as CPython3 (the default in Revit 2024). Right-click a Python node and check the engine.

Sightline Schematic 2026.11.05.03. Guide file: Stadium_Bowl_Mass_Revit_R2024_Guide.pdf. Menu names are for SketchUp 2024 and Revit 2024 (Dynamo 2.17); other versions may differ slightly.